

Supplementary Material S1: Notation Reference

Supplementary Material S1: Notation Reference I

This supplement provides a compact reference for all mathematical notation used in this paper. Definitions originate from Paper 1 [?] unless noted otherwise.

Cognitive State Notation (Paper 1) I

Symbol	Name	Definition
σ_i	Cognitive state of agent i	$\sigma_i = \langle \mathcal{B}_i, \mathcal{G}_i, \mathcal{J}_i, \mathcal{H}_i \rangle$
$\mathcal{B}_i$	Belief set	Probability distribution over propositions
$\mathcal{G}_i$	Goal set	Prioritized objectives with utility weights
$\mathcal{J}_i$	Intention set	Committed action plans
$\mathcal{H}_i$	History	Interaction trace (messages, actions, observations)
Σ	System state	$\Sigma = \{\sigma_1, \dots, \sigma_n\}$ for n agents

Trust Calculus (Paper 1) I

Symbol	Name	Definition
$\mathcal{T}_{i \rightarrow j}$	Trust from agent i to agent j	$\mathcal{T}_{i \rightarrow j}^t = \alpha \cdot T_{\text{base}}(j) + \beta \cdot T_{\text{rep}}^t(j) + \gamma \cdot T_{\text{ctx}}^t(i, j)$
T_{base}	Architectural trust	Role-based, static assignment
T_{rep}	Reputation trust	Historical accuracy, time-decayed
T_{ctx}	Context trust	Task-specific, situational
δ	Delegation decay factor	$\delta \in (0, 1)$; trust decays as δ^d over delegation depth d

Defense Mechanism Notation (Paper 1) I

Symbol	Name	Definition
$\mathcal{F}(m)$	Cognitive Firewall	$\mathcal{F}(m) \rightarrow \{\text{accept, quarantine, reject}\}$
τ	Firewall threshold	Trust score cutoff for accept/reject decision
$\mathcal{B}_{\text{verified}}$	Verified belief partition	Beliefs promoted through corroboration protocol
$\mathcal{B}_{\text{provisional}}$	Provisional belief partition	Sandboxed beliefs awaiting verification
$\mathcal{W}$	Canary belief set (Tripwires)	Sentinel beliefs that trigger alerts if modified
INV_k	Behavioral invariant k	Runtime predicate: $\text{INV}_k(\sigma_i) \in \{\text{true, false}\}$
S_{drift}	Drift detection score	$S_{\text{drift}} = (\mathcal{B}_i^t \ \mathcal{B}_i^{t-1})$
ϵ	Drift threshold	Maximum tolerable KL divergence

Defense Mechanism Notation (Paper 1) II

$\mathcal{B}_{\text{consensus}}$	Byzantine consensus belief	Agreed-upon belief across quorum q of agents
q	Quorum size	Minimum agents required for consensus; $n \geq 3f + 1$
f	Byzantine fault tolerance	Maximum number of compromised agents tolerated

Adversary Taxonomy (Paper 1) I

Symbol	Class	Scope	Access
Ω_1	External	User boundary	Direct prompt manipulation
Ω_2	Peripheral	Data/tool channels	Indirect injection via data poisoning
Ω_3	Agent-level	Single agent	Compromised agent with modified goals
Ω_4	Coordination	Inter-agent	Man-in-the-middle on agent communication
Ω_5	Systemic	Orchestrator	Full framework-level compromise

Axiomatic Design Notation (This Paper) I

Symbol	Name	Definition
$\{FR\}$	Functional Requirements vector	Objectives the system must satisfy
$\{DP\}$	Design Parameters vector	Variables chosen to satisfy FRs
$[A]$	Design Matrix	Maps DPs to FRs: $\{FR\} = [A]\{DP\}$
A_{ij}	Matrix element	Coupling coefficient: $\partial FR_i / \partial DP_j$
$[A']$	Coupled Design Matrix	Post-attack matrix with off-diagonal terms introduced by adversary

Axiomatic Design Notation (This Paper) II

Design Matrix States:

- ▶ **Uncoupled** (diagonal $[A]$): Each FR depends on exactly one DP. Independence Axiom satisfied.
- ▶ **Decoupled** (triangular $[A]$): FRs can be satisfied sequentially. Acceptable but fragile.
- ▶ **Coupled** ($[A']$ with off-diagonal terms): FRs interfere. Adversarial transient coupling makes the system unstable.

OODA Loop Notation (This Paper) I

Phase	Function	CIF Attack Surface
Observe	Sense environment, ingest data	Data channel integrity (sensors, APIs, logs)
Orient	Synthesize observations with prior knowledge	Primary target of Goal Hijacking: internal model corruption
Decide	Select action based on oriented model	Action space restriction, capability elicitation
Act	Execute selected action	Unauthorized action execution

OODA Phase $\leftrightarrow$ CIF Defense Mapping I

OODA Phase	Primary CIF Defense	Mechanism
Observe	Cognitive Firewall ($\mathcal{F}$)	Filter/classify incoming data before it reaches Orientation
Orient	Belief Sandboxing ($\mathcal{B}_{\text{provisional}}$), Drift Detection (S_{drift})	Isolate new beliefs; detect sudden orientation shifts
Decide	Behavioral Invariants (INV_k)	Verify decisions against pre-defined safety predicates

OODA Phase $\leftrightarrow$ CIF Defense Mapping II

Act	Byzantine Consensus ($\mathcal{B}_{\text{consensus}}$)	Require multi-agent agreement before critical actions
All phases	Behavioral Invariants (INV_k)	Continuous runtime monitoring across the full cycle

Universal Attack Patterns (This Paper) I

Pattern	Description	Design Matrix Effect
FR Polarity Inversion	Adversary flips a negative FR (minimize cost) to positive (maximize output of harmful byproduct)	Diagonal element A_{ii} changes sign
Constraint Relaxation	Hard safety constraint degraded to soft preference	Diagonal element A_{ii} reduced toward zero

Universal Attack Patterns (This Paper) II

Context Boundary Violation

Isolated
operational
contexts bleed
together (e.g.,
simulation $\rightarrow$
operational)

Off-diagonal element A_{ij}
introduced where $i \neq j$
